INTELLIGENT DECISION SUPPORT SYSTEM FOR SRI LANKAN TEA ESTATE MANAGEMENT.

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Labor Management and Welfare Optimization System

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i

ABSTRACT

The management of workforce at Sri Lankan tea estates relies largely on labor management methods that are manually conducted. Such a situation may create certain difficulties when it comes to the monitoring of the workers' welfare and labor management decision-making. The workers conduct physiologically challenging tasks under different working conditions, while the manager needs to get timely information about their welfare and manage the workforce efficiently. This research introduces the Labor Management and Welfare Optimization System as one of the components of the Intelligent Decision Support System for Sri Lankan Tea Estate Management.

This component uses machine learning, Internet of Things (IoT) and web technologies for the purposes of the monitoring of workers' welfare and labor management. Data about the workers' physiology, such as Heart Rate (HR) and Blood Oxygen Saturation (SpO_2), obtained through IoT devices is used for identification of their state. The PhysioNet Non-EEG Dataset for Assessment of Neurological Status was used for training the LightGBM Classifier (LGBMClassifier) based on HR and SpO_2. This model classifies the states of workers into the following four classes: Relaxed, Emotional Stress, Cognitive Stress and Physical Stress. The accuracy of the model is 82%.

The model was embedded in a web application in which the received real-time physiological data is processed and the worker's state is displayed in the dashboard. This component also provides the functionality of worker's information management, health monitoring and task assignment suggestion that helps the estate managers make labor decisions.

Keywords: Labor Management, Worker Welfare, Internet of Things, LightGBM, Machine Learning

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DL	Deep Learning
HR	Heart Rate
IoT	Internet of Things
LGBM	Light Gradient Boosting Machine
ML	Machine Learning
NLP	Natural Language Processing
SDG	Sustainable Development Goal
SpO_2	Blood Oxygen Saturation
UI	User Interface
UX	User Experience

1 INTRODUCTION

1.1 Overall project context

The tea industry is an important agricultural sector in Sri Lanka and provides employment and livelihoods for many rural communities. Effective labour management is essential for maintaining productivity and ensuring the welfare of tea estate workers. However, traditional workforce management approaches may provide limited information about workers' conditions during daily field activities.

Recent developments in Machine Learning (ML) and Internet of Things (IoT) provide opportunities to improve worker monitoring and labour management. IoT wearable devices can collect physiological information, while machine learning can analyse this information to provide useful predictions for management decisions.

The proposed Labor Management and Welfare Optimization System applies these technologies to support worker monitoring and welfare. The system uses Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) collected through IoT wearable devices and analyses them using a LightGBM Classifier. The model classifies worker conditions into Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress categories and achieved an accuracy of 82%.

The predicted worker condition is displayed through a web-based dashboard, allowing estate managers to monitor workers and identify those who may require attention. This provides a data-driven approach to supporting labour management, worker safety, and welfare.

1.2 Individual component context

The Labor Management and Welfare Optimization System is the individual component developed to support better workforce management and worker welfare in Sri Lankan tea estates. The component focuses on monitoring the physiological condition of workers during field activities and providing useful information to estate managers through an intelligent, web-based system.

The system uses IoT wearable devices to collect workers' physiological data. Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) are used as the main input features

because they provide information about the physical condition of workers during their activities.

For intelligent analysis, the PhysioNet Non-EEG Dataset for Assessment of Neurological Status was used to train a LightGBM Classifier (LGBMClassifier). The model classifies worker conditions into four categories: Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress. The trained model achieved an accuracy of 82%.

The trained model is integrated with the web application. During operation, real-time HR and SpO_2 values collected from the IoT device are sent to the prediction model, and the predicted worker condition is displayed on the dashboard. This enables estate managers to monitor worker conditions and identify workers who may require attention during daily activities.

Therefore, the individual component combines IoT-based physiological monitoring, machine learning-based worker condition classification, and web-based dashboard monitoring to support labour management, worker safety, and welfare in tea estate environments.

1.3 Problem statement

Labour management is an important aspect of tea estate operations because the performance of field activities depends heavily on workers. However, monitoring the condition of workers during daily activities can be challenging when managers rely mainly on conventional observation and management practices. This can make it difficult to obtain timely information about a worker's physical condition or stress level while the worker is engaged in field activities.

Existing research has explored machine learning for workforce management, productivity analysis, and worker condition prediction [2], [4], [5]. However, the research paper identifies the need to combine workforce management with continuous worker health monitoring. In particular, conventional labour management approaches do not provide a continuous mechanism for collecting physiological information from workers and using that information to support management decisions.

Therefore, the main problem addressed by this research is the lack of an intelligent and continuous approach for monitoring tea estate workers' physiological conditions while supporting labour management. Without such information, estate managers may have limited ability to identify workers experiencing stress or conditions that may require attention during field activities.

To address this problem, the proposed system integrates IoT wearable devices for collecting Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) with a LightGBM machine learning classifier. The system predicts worker conditions as Relaxed, Emotional Stress, Cognitive Stress, or Physical Stress and presents the prediction through a web-based dashboard. 41c8cfc7-01c6-4341-884e-8d69184…

Thus, this research focuses on developing a data-driven solution that can provide real-time physiological information and machine learning-based predictions to support labour management, worker safety, and welfare in Sri Lankan tea estates.

1.4 Research gap

Previous research has explored the use of Machine Learning (ML) for workforce management, productivity analysis, worker fatigue prediction, and intelligent task allocation [2], [4], [5]. These studies demonstrate that intelligent technologies can support workforce-related decision making and improve the analysis of worker-related information.

However, the existing approaches identified in the research do not sufficiently address the combination of labour management and continuous worker health monitoring within the Sri Lankan tea estate context. The research paper identifies that existing labour-related studies can support labour prediction, task assignment, and productivity improvement, but they do not adequately monitor the health condition of workers while they are performing their activities.

In addition, Sri Lankan tea estate labour involves physically demanding field activities, making worker welfare an important consideration alongside workforce management. Existing approaches therefore leave a gap in providing an integrated mechanism that can collect workers' physiological information and use machine learning to identify their conditions during daily activities.

Therefore, the research gap addressed by this study is the limited integration of IoT-based real-time physiological monitoring with machine learning-based worker condition classification for supporting labour management and worker welfare in Sri Lankan tea estates. The proposed component addresses this gap by combining wearable physiological monitoring with machine learning and web-based monitoring within a single system.

1.5 Aim and objectives

1.5.1 Aim

I feel the main aim of this research is to create a Labor Management and Welfare Optimization System, for Sri Lankan tea estates.The Labor Management and Welfare Optimization System will use based physiological monitoring and machine learning.The Labor Management and Welfare Optimization System aims to help monitor worker condition manage labour and improve worker welfare.

1.5.2 Objectives

1) Develop an IoT-based wearable monitoring approach for collecting workers' physiological data, particularly Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) during field activities.
2) Develop a machine learning-based worker condition classification model using the PhysioNet Non-EEG Dataset for Assessment of Neurological Status.
3) Implement a LightGBM Classifier (LGBMClassifier) to classify worker conditions into Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress categories.
4) Integrate the trained machine learning model with the web-based application so that real-time physiological data can be processed, and worker conditions can be displayed through the dashboard.
5) Evaluate the performance of the developed worker condition classification model and assess its potential to support labour management, worker safety, and welfare in tea estate environments.

1.6 Research questions or hypotheses

RQ1 : How can IoT wearable devices collect physiological data, Heart Rate (HR) and Blood Oxygen Saturation (SpO_2), from tea estate workers in a field?

RQ2 : How effective is the use of the Machine Learning algorithm to classify the tea estate workers' condition based on the collected data.

RQ3 : How efficient is the use of the LightGBM Classifier to classify workers into Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress conditions?

RQ4 : How can the proposed machine learning solution be embedded within a web-based system to manage workers' conditions?

RQ5 : How effective is the system in supporting the tea estate managers in monitoring and identifying the workers' conditions.

Hypotheses

H_0 (Null Hypothesis): The integration of IoT wearable devices and Machine Learning algorithms is not an effective solution to manage tea estate workers' conditions.

H_1 (Alternative Hypothesis): The integration of IoT wearable devices and Machine Learning algorithms is an effective solution to manage tea estate workers' conditions.

1.7 Scope limitations and assumptions

Scope

The component of the overall Intelligent Decision Support System for Sri Lankan Tea Estate Management that focuses on Labour Management and Welfare Optimization has the objective of monitoring the conditions of the workers and assisting in labour management decisions based on IoT-based physiological monitoring and machine learning.

• Collecting the physiological data of the workers using IoT wearable devices.

• Monitoring Heart Rate (HR) and Blood Oxygen Saturation (SpO_2).

• Processing the physiological data collected.

• Using the LightGBM Classifier (LGBMClassifier) for worker condition classification.

• Classifying the worker conditions into Relaxed, Emotional Stress type, Cognitive Stress type, and Physical Stress type.

• Displaying the predicted worker condition using web application dashboard.

• Assisting estate managers with monitoring the conditions of the workers during their activities in the fields.

• Integrating the worker monitoring and prediction functionality into the overall decision-support platform.

Limitations

First of all, the quality and appropriateness of the data sets used for the classification model training and validation by the worker condition classification model greatly impact its performance. The LightGBM classifier was trained using PhysioNet Non-EEG Dataset for Assessment of Neurological Status, and it achieved an accuracy of 82%.

Secondly, the model uses two vital signs, namely, Heart rate and Oxygen saturation, as the input parameters. However, there are several other factors that can affect a person's condition, for instance, age, a person's physical state, the amount of work they have to do, and the environment they have to work in.

Another limitation is that the model was trained on a general dataset and thus might not account for all the variations specific to the Sri Lankan tea estate workers, as a specific data set was not compiled for the task. Furthermore, the predicted worker condition should not be used as a substitute for a professional medical examination. The model's results can be used as a decision support system for the tea estate managers to keep track of their workers' conditions and ensure their well-being.

Assumptions

IoT wearable devices will provide reliable Heart Rate (HR) and Blood Oxygenation (SpO2) readings during the workers' daily work.

The physiological data received from the wearable devices will be delivered to the web application, where it will be processed by the trained algorithm, thus predicting workers' condition.

The study assumes that tea estate managers will have access to the web dashboard, where the predicted data will be presented and read. It is believed that the information will help the managers monitor the workers' well-being and detect workers needing immediate assistance. However, the algorithm's results will only be used as a decision-support tool and cannot replace medical expertise.

1.8 Individual Contribution

This thesis presents the individual contributions of Gunasekara L.M.N.P. (IT22267986) towards the research project R26-IT-118 entitled "Intelligent Decision Support System for Sri Lankan Tea Estate Management." This research project encompasses four major modules and the individual contribution is focused on the Labour Management and Welfare Optimization System.

The individual contribution involved in the project includes the development of a labor management and worker welfare monitoring component using machine learning and Internet of Things (IoT) technologies. Further, it encompasses the research on monitoring the worker condition, review of literature, identification of suitable physiological input features, preparation of the dataset, and design of the machine learning-based solution.

In the machine learning-based solution, a LightGBM Classifier (LGBMClassifier) is utilized in the project for classifying the worker conditions based on Heart Rate (HR) and Blood Oxygen Saturation (SpO_2). The PhysioNet Non-EEG Dataset for Assessment of Neurological Status was used to train the model. The trained model classifies the worker's condition based on four categories, namely, Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress. The accuracy attained by the trained model was 82%.

The contribution also integrates the trained machine learning-based model into the web application. The application consists of React.js as the frontend framework,

Node.js and Express.js for running the server-side, and a MongoDB database for storing the data. The physiological data obtained from IoT wearable smart devices are applied to the prediction model where the predicted worker's condition is displayed in the web dashboard for the estate managements.

One additional individual contribution to the project is to support labour management and worker welfare by presenting worker condition predictions. By presenting the worker's predicted conditions, the system aims to provide an insight into the condition of the workers during field activities. The contribution contributes towards worker welfare in the Sri Lankan tea estates by considering their physiological conditions and utilizing them in a machine learning model for classification and prediction tasks. The application of the prediction results in a web dashboard provides an intuitive understanding of the worker's condition for estate managements.

1.9 Thesis Structure

Chapter 1 – Introduction

This chapter introduces the overall TeaGuard research project and the individual Labor Management and Welfare Optimization component. The research problem, research gap, aim, objectives, research questions, scope, limitations, assumptions and individual contribution will be explained in this chapter.

Chapter 2 – Literature Review

This chapter reviews the relevant existing research in relation to the Labour Management and Welfare Optimization project. Previous research, worker stress and fatigue prediction based on physiological indicators, and related IoT based monitoring systems will be examined. The limitations of the existing approaches and the research gap addressed by this research will also be explained.

Chapter 3 – Research Methodology

This chapter explains the methodological approach that will be adopted in developing and evaluating the research component. The research approach, dataset

selection, data preparation, feature selection, machine learning model design, model training and evaluation will be discussed.

Chapter 4 – Product and Solution Design

The design of the Labor Management and Welfare Optimization Solution will be discussed in this chapter. The system overall architecture, IoT wearable approach, physiological data acquisition and processing, machine learning powered prediction workflow, backend communication, database integration and dashboard design will be explained.

Chapter 5 – Implementation, Integration and Testing

This chapter will discuss the implementation approach of the individual component and system level integration. The technologies selected, machine learning model implementation, front-end and back-end implementation, database implementation, IoT data processing, and dashboard implementation will be discussed. The integration of the different elements and testing procedures will be presented as well.

Chapter 6 – Results and Validation

The results of the developed worker condition classification model and system level implementation will be discussed and evaluated in this chapter. The performance of the model, including an expected 82% accuracy will be presented.

Chapter 7 – Discussion

The discussion of results in relation to the research objectives and findings of previous research will be presented in this chapter. The effectiveness of the developed solution, contributions to the field of worker condition monitoring, and its potential benefits to labour management and worker welfare in Sri Lankan tea estates will be explored. The limitations and challenges of the developed component will also be discussed.

Chapter 8 – Commercialization and Sustainability

This chapter considers the commercialization potential and practical applications of the component. Possible users, including tea estate managers and estate management organisations, and the potential value of the system for worker monitoring and

welfare management will be explored. The feasibility of future deployment, maintenance requirements and opportunities for future development will also be considered.

Chapter 9 – Conclusion and Future Work

The final chapter summarises the research, presenting the extent to which the research aim and objectives have been achieved, and the main contributions of the Labor Management and Welfare Optimization component. The limitations of the solution and possible future improvements, such as further model evaluation and enhancement of physiological monitoring, will also be presented.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The literature review of this thesis focuses on related research on labour management, worker welfare, machine learning-worker condition classification, and internet of things-physiological monitoring. The objective of this review is to identify the opportunities to apply intelligent systems in order to contribute to the efficient management and enhanced welfare of workers in Sri Lankan tea estates.

The review covers prior research on application of machine learning techniques in workforce management, productivity analysis, worker fatigue prediction, and intelligent workforce scheduling. Such analysis serves as the basis for decision-making in workforce management while taking into account the benefits that can be derived from implementing machine learning techniques [2], [4], [5].

The review also focuses on the importance of ensuring worker welfare while managing agricultural labour forces, considering the strenuous working conditions in tea plantations. Monitoring and evaluating the physiological conditions of the tea workers can benefit the management process in tea plantations, thus the interest in applying internet of things enabled wearable sensors to continuously record relevant physiological metrics, such as the Heart Rate (HR) and Blood Oxygen Saturation (SpO_2), for developing a machine learning-worker condition classification model.

Besides that, the review focuses on the application of classification algorithms and their role in worker condition prediction, which serves as a basis for decision-making in the proposed Labor Management and Welfare Optimization System. In the proposed system, an optimized Light Gradient Boosting Machine Classifier (LGBMClassifier) was trained using the PhysioNet Non-EEG Dataset for Assessment of Neurological Status to predict the worker's condition (Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress). The predicted condition is then displayed using a web dashboard.

The key areas covered by this literature review include:

1. Application of machine learning techniques in workforce management and productivity analysis;

2. Application of machine learning techniques in worker fatigue, stress, and physiological condition prediction;

3. Physiological monitoring using internet of things-enabled wearable sensors;

4. Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) as key features for implementing the worker condition prediction model;

5. Application of classification algorithms for worker condition prediction;

6. Light Gradient Boosting Machine classification and its implementation in the proposed system;

7. Overview of web dashboard implementation to monitor the worker's condition;

8. Limitations of workforce management and worker monitoring techniques;

9. Research gap that the proposed Labour Management and Welfare Optimization System aims to address.

Thus, the scope of this review is limited to published works that directly relate to the development of an effective labour management component of the overall TeaGuard research project. The review focuses on identifying the opportunities to implement physiological data acquisition, machine learning-worker condition classification, and web dashboard implementation to manage labour forces and enhance the welfare of the tea workers in Sri Lankan tea estates.

2.2 Domain and theoretical foundations

2.2.1 Labour Management in Tea Estates

Labour management is an essential aspect of tea estate management since most activities within these enterprises are highly dependent on the characteristics of the workforce. It is necessary to ensure the proper organisation of processes and the

provision of individual worker care. At the same time, labour management in commercial enterprises is a challenging task, particularly in agriculture, where employees work in open spaces. Thus, developing innovative solutions for estimating the physiological state of a worker in real-time during field activities is extremely relevant.

Workforce management practices traditionally rely on observational data and managerial expertise, which can help in day-to-day decision-making but offer little help in analysing the workers' physiological condition. As a result, intelligent systems can complement traditional methods by providing closed-loop systems with additional data about the employees' health. A personal research component is the optimization of labor management and welfare based on the principles of physiological information analysis and machine learning-assisted worker condition classification.

2.2.2 Artificial Intelligence and Machine Learning in Workforce Management

Artificial Intelligence (AI) refers to the ability of a computer system to carry out activities that require human intellect to analyze data, identify patterns, and make decisions. Machine Learning (ML) is a branch of AI that involves the use of algorithms to enable computers to undertake tasks or operations without human intervention. ML uses data to train algorithms and then makes decisions or predictions based on the information.

In workforce management, ML algorithms can process data to classify and analyze workforce productivity and make other decisions related to the workforce. Earlier works have demonstrated the use of ML algorithms for workforce management, productivity analysis, and prediction of worker fatigue and intelligent scheduling [2], [4], [5].

In the proposed component, ML algorithms can be used to classify the data collected from the workers' physiological data acquisition devices. These algorithms can help analyze the information and present the predicted status to the estate managers in a web dashboard.

2.2.3 Physiological Monitoring of Workers

Physiological monitoring includes the collection and analysis of information that identifies the state of a physical body. Monitoring wearable devices can gather the data during day-to-day activities and be processed by a computerized algorithm.

Heart Rate (HR) and Blood Oxygenation level (SpO2) are selected for the Labor Management and Welfare Optimization System as the input parameters for worker's condition classification. IoT devices enable the physiological measurements, and the data is utilized in the machine learning model for processing.

The use of physiological measurements in the system can add valuable information for worker surveillance during field activities. However, relying on these two parameters cannot ensure the identification of a person's health status or potential disease diagnosis. It should be noted that the system's predictions are applied to monitor and control workers and should not be considered as an alternative to medical investigation.

2.2.4 Internet of Things and Wearable Devices

The Internet of Things or IoT is a network of physical devices that gather and share data. IoT technology brings together sensors, communication systems and software applications to allow the gathering and processing of information.

Wearable devices can be used in IoT-based monitoring systems to gather measurements from people. In the proposed component wearable devices give SpO_2 measurements that are used for worker condition classification. The gathered information is sent to the application. Processed by the trained machine learning model.

Integrating monitoring with a web-based application gives a way to make physiological information available, to estate managers. This approach helps build a monitoring system that combines data collection, automatic analysis and dashboard-based presentation.

2.2.5 Heart Rate and Blood Oxygen Saturation

Heart Rate (HR) is the number of heartbeats per minute. The HR may vary depending on physical activities, emotional reactions, etc. Thus, the heart rate measurements may serve as physiological data in monitoring individuals while doing their everyday activities.

Blood Oxygen Saturation (SpO_2) is the calculated percentage of haemoglobin bound with oxygen in the blood. It can be measured with the help of adequate optical sensors. The accuracy of measurement may depend on a number of factors including the quality of devices, movements, etc.

The created worker condition classifier works on HR and SpO_2 data as the primary features. This data is classified by the machine learning algorithm into one of four conditions: Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress.

2.2.6 Worker Condition Classification Using Machine Learning

Classification is a machine learning technique that involves supervised learning from labelled instances, wherein the model is trained to assign an instance to one of a predetermined number of classes. The model is trained with a dataset comprising the input and the class labels of the input.

The HR and SpO_2 were used as the input variables to classify the conditions of the workers in this component. The model was trained using the PhysioNet Non-EEG Dataset for Assessment of Neurological Status. The task involved four classes, which are Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress.

The trained model was able to perform the task with an accuracy of 82%. The aforementioned figure denotes the classification accuracy of the model during its testing in the given conditions in the research. The model's performance in the actual Sri Lankan tea estates may need validation.

2.2.7 LightGBM Classifier

LightGBM is a gradient boosting framework for machine learning tasks such as classification and regression. It consists of multiple decision trees built to form an

ensemble that learns the connection between the input data features and the target output labels. The framework is also applicable for solving regression tasks.

The Labor Management and Welfare Optimization System employs the LightGBM Classifier, specifically the "LGBMClassifier" module, for classifying the physiologic input data into the predicted worker condition categories. The classifier was trained to yield the accuracy score of 82% on the selected PhysioNet dataset.

It was then embedded into the application's web-based interface, where the physiologic data from the workers are inputted and processed to display the predicted worker condition in a dashboard. This way, the process of analyzing the workers' physiological state becomes reliant on the machine learning model.

2.2.8 Web-Based Decision-Support Systems

A web-based decision-support system is a system that provides information that can help an entity make informed decisions. Such a system typically involves a combination of a user interface, backend services, data storages, and processing components that collect, manage, and display the information.

The individual component is designed as a web application built with React.js on the frontend, Node.js and Express.js on the backend, and MongoDB as the database to store the information. The application allows worker information management, physiological condition monitoring, and predicted worker condition presentation.

Thus, when physiological measurements are entered into the prediction model, the resulting worker condition is presented on the web-based dashboard. Thus, with the proposed system, estate managers can utilise the available monitoring information and functionality in a single application to identify the workers in need of further attention.

2.2.9 Integration of Physiological Monitoring and Machine Learning

Integration of IoT-based physiological monitoring and an approach using machine learning provides an opportunity to combine data collection and analysis. As such, the wearable device collects physiological measurements, and the classification model analyses the input data and provides a predicted worker condition.

The general workflow of the individual component can be presented in the following manner:

Such a workflow combines data collection from physiological sensors, machine learning-based analysis and presentation of the results in a web dashboard. The predicted condition can provide decision-support information which can help the estate managers to monitor the workers during the field activities.

In terms of the developed component, an accuracy of data classification was reported as 82%. However, the usefulness of such a system in operational tea estates is conditional to the reliability of wearable measures, successful data transmission and further validation under working conditions.

The general workflow of the individual component can be represented as follows:

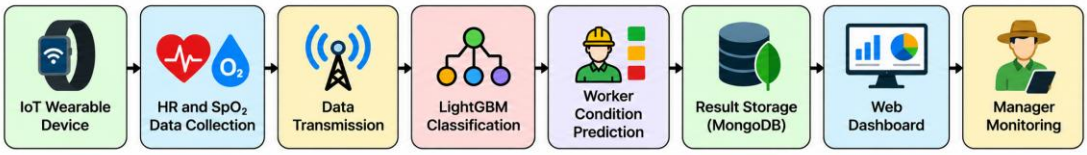


Figure 2.1: Labor Management and Welfare Monitoring Workflow

3 RESEARCH METHODOLOGY

3.1 Research design

The study has been designed in accordance with a design and development research methodology to evolve the Labor Management and Welfare Optimization component of Intelligent Decision Support System for Sri Lankan Tea Estate Management. The component evolved utilizes Internet of Things technology in conjunction with Machine Learning to monitor the welfare of the workers.

The research work focuses on evolving a worker condition classification framework based on physiological data collected from IoT-enabled wearables. The LightGBM Classifier (LGBMClassifier) has been utilized for classifying worker conditions based on Heart Rate (HR) and Blood Oxygen Saturation (SpO_2). The PhysioNet Non-EEG Dataset for Assessment of Neurological Status has been utilized in this work to classify the worker conditions into one of the four classes namely, Relaxed, Emotional Stress, Cognitive Stress and Physical Stress.

The study involves identifying the problem, selecting and preparing the dataset, developing the model, evaluating the model, integrating the model into the system and validating the same. Table 3.1 below depicts the various research work involved and its association with the thesis.

Table 3.1: Research activities and their relationship to the study

Research activity	**Activity in this study**	**Thesis chapter**
Problem identification	Identifying the challenges of monitoring workers' physiological conditions and supporting worker welfare in tea estates	1, 2
Define objectives	Developing a machine learning-based worker condition monitoring system	1
Literature review	Reviewing labour management, worker welfare, IoT-based physiological monitoring, and machine	2

	learning techniques	
Design and development	Dataset preparation, LightGBM Classifier development, and web application integration	3, 4, 5
Demonstration	Demonstrating worker condition classification and presenting predictions through the web-based dashboard	5, 6
Evaluation	Evaluating the worker condition classification model, which achieved a reported accuracy of 82%	3, 6
Communication	Preparing the final thesis, system documentation, and research findings	All

The activities described in Table 3.1 below represent the stages involved in development of the Labor Management and Welfare Optimization component. These activities include identifying the research problem, reviewing the existing literature, preparing the dataset, developing and evaluating the machine learning model, and integrating the developed model with the web application. The last activity involved validating the implemented system and documenting the research results.

3.2 Data participants or study materials

3.2.1 Physiological Dataset

The primary material used in this research is the PhysioNet Non-EEG Dataset for Assessment of Neurological Status. This dataset is used to develop the worker condition classification model in the Labor Management and Welfare Optimization component of the Intelligent Decision Support System for Sri Lankan Tea Estate Management.

The dataset is used to train the LightGBM Classifier (LGBMClassifier) using physiological information. Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) are

the main input features used by the model to classify worker conditions into four categories: Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress.

Finally, the dataset is used for model development and evaluation. The trained model is integrated into the web application to present worker condition predictions through the dashboard.

Table 3.2: Main study materials

Study material	**Purpose**
PhysioNet Non-EEG Dataset for Assessment of Neurological Status	Training and evaluating the worker condition classification model
Heart Rate (HR)	Providing physiological input for worker condition classification
Blood Oxygen Saturation (SpO_2)	Providing physiological input for worker condition classification
LightGBM Classifier (LGBMClassifier)	Classifying worker conditions
IoT wearable devices	Collecting physiological measurements
React.js	Developing the frontend and dashboard
Node.js and Express.js	Backend processing and API communication
MongoDB	Storing application data
Worker condition predictions	Presenting classification results through the dashboard

3.2.2 Study Participants

The main model development process pulls data from the PhysioNet Non-EEG Dataset for Assessment of Neurological Status. Because of that the model training experiment relies on the chosen dataset of gathering new data directly from tea estate workers.

The developed system is made for people involved in tea estate management, estate managers who need to keep track of workers' conditions. Those users can use the web application to see worker information and predicted conditions shown on the dashboard.

Any future evaluation that involves tea estate workers must happen apart, from the model training experiment. That evaluation must follow ethical rules and gather only the data needed to test the system.

3.3 Experimental or Study Procedure

1. Physiological dataset selection and preparation.
2. Selection of Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) features.
3. LightGBM Classifier model development and training.
4. Model evaluation and performance assessment.
5. Web application integration.
6. Worker condition prediction.
7. Prediction result presentation.
8. Database storage and dashboard display.

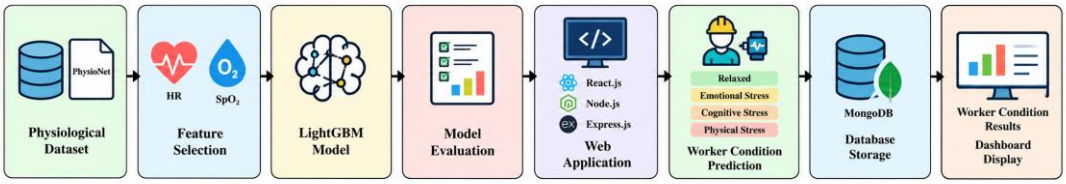


Figure 3.1: Experimental procedure for Labor Management and Welfare Optimization System

Figure 3.1: Experimental Procedure for Labor Management and Welfare Optimization System.

3.3.1 Dataset Preparation

This step involves preparing a dataset for implementation using PhysioNet Non-EEG Dataset for Assessment of Neurological Status benchmark. This will be used to train and test the best performing model, LightGBM Classifier (LGBMClassifier).

Heart Rate (HR) and Blood Oxygenation (SpO2%) are the physiological attributes used in this study for worker condition classification. The dataset serves as the basis for the worker condition classification model, while the IoT wearable device provides real-time worker physiological measurements for the system.

The created dataset will be used to test the worker condition classification model. During deployment, the IoT wearable device captures measurements which are then processed by the trained model to predict worker condition.

3.3.2 Model Development and Training

The constructed database will be applied for the development and training of the Light Gradient Boosting Machine Classifier (LGBMClassifier). This algorithm will adopt the physiological input data and train on them to enable the system to identify worker conditions based on the acquired features, namely, Heart Rate (HR) and Blood Oxygen Saturation (SpO_2).

The classification model will comprise four categories of worker conditions, including Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress. The constructed classification model will be employed in the developed system to predict worker conditions once the physiological input parameters are obtained via the IoT wearable device.

3.3.3 Model Evaluation

The Classifier (LGBMClassifier) was used to evaluate how well it could classify worker conditions based on physiological data. The model was trained using the PhysioNet Non-EEG Dataset for Assessment of Neurological Status. The main input features used were Heart Rate (HR) and Blood Oxygen Saturation (SpO_2).

The model sorts worker conditions into four groups: Relaxed, Emotional Stress, Cognitive Stress and Physical Stress. According to the results in the research paper the trained model reached an accuracy of 82%.

After training the classifier was added to the web-based application. When the system runs, the Heart Rate (HR). Blood Oxygen Saturation (SpO_2) readings from the IoT wearable device are sent to the prediction model. The model then predicts the worker's condition. The result appears on the dashboard. This helps estate managers keep track of their workers during tasks.

The accuracy reported here reflects the model's performance, under the conditions of the study. To better understand how the model works in real-world tea estate settings more testing is needed using data collected directly from tea estate workers.

3.3.4 IoT: Data collection and communication

IoT is designed to collect physiological measurements of the worker while performing a daily task. The physical activity of a given worker is captured and relayed to the back end. An ESP32 microcontroller and a MAX30102 sensor were employed to capture Heart Rate (HR) and Blood Oxygenation (SpO2). At the same time, the worker was requested to provide a finger print on the MAX30102 sensor for the collection of the physiological metrics.

ESP32 was configured to send the collected information to the backend using the Message Queuing Telemetry Transport (MQTT). The data sent by the IoT device are received by the MQTT broker and stored for processing by the Node.js backend.

The backend receives the data and stores the required data in the MongoDB. It is also the backend that collects the data samples that were captured within a 30-second window. From the samples, features were extracted and used to train the LightGBM Classifier. The classifier makes it possible to predict the condition of the worker based on the features of the machine learning model.

The results of the prediction are then used to recommend or suggest the tasks to the workers. The dashboard is also used by the estate managers to collect the information needed to make decisions about the distribution of tasks.

3.3.5 Feature Extraction and Worker Condition Prediction

The process of predicting the worker's condition is based on the physiological data acquired by the IoT wearable device. Specifically, two channels, including Heart Rate (HR) and Blood Oxygenation (Saturation), are utilized by the Light Gradient Boosting Machine (LGBMC) algorithm.

Prior to making a prediction, a set of extracted statistical features from the given 30 seconds of the input data is used as an input for the trained classifier. The following variables, which represent the mean, maximum, minimum, and standard deviation values of the HR and saturation levels, along with the number of saturation drops, were used for feature engineering.

The LGBMC model was trained on the PhysioNet Non-EEG Dataset for Assessment of Neurological Status and showed an accuracy of 82%. The algorithm classifies the worker's condition into four categories: Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress.

Finally, the model's outputs are used to provide an update on the workers' health status and recommend certain tasks based on their condition. The results of the prediction are available in the web application, which helps the estate manager analyze the situation and make appropriate decisions.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders

The principal stakeholders for the Labor Management and Welfare Optimization module include people who come into contact with the system or are affected by the processes carried out by the system. The following is a summary of the principal stakeholders in Table 4.1.

Table 4.1: Stakeholders and Their Roles and Interests

Stakeholder	Role and Interest
Estate Manager	Registers workers, monitors worker conditions, reviews task recommendations, and manages workforce information.
Tea Estate Workers	Provide physiological measurements through the IoT monitoring device and benefit from improved worker welfare monitoring.
Supervisor	Reviews recommended tasks, approves assignments, and oversees the allocation of work.
System Administrator	Maintains the application, manages system access, and supports the technical operation of the system.
Researcher / Developer	Develops and maintains the machine learning model, IoT integration, backend services, database, and dashboard.

4.1.2 Functional Requirements

The functional requirements specify the core functions of the Labor Management and Welfare Optimization System. These include worker enrollment, data collection, predicting worker conditions, task suggestions, supervisor authorization, and dashboard display.

Table 4.2 : Functional Requirements

Functional Requirement	**Description**
Worker Registration	The system shall allow the estate manager to register workers and maintain their basic details.
Physiological Data Collection	The system shall collect Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) measurements using the MAX30102 sensor connected to an ESP32.
Data Transmission	The system shall transmit sensor measurements through MQTT using the HiveMQ broker.
Data Validation	The backend shall validate received physiological measurements before further processing.
Data Storage	The system shall store worker information and relevant monitoring records in MongoDB.
Feature Extraction	The system shall extract statistical features from physiological measurements collected over a 30-second window.
Worker Condition Prediction	The system shall use the trained LightGBM Classifier to classify worker conditions into Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress.
Task Recommendation	The system shall provide task recommendations according to the defined worker risk-management rules.
Supervisor Approval	The system shall allow the responsible supervisor to review and approve recommended task assignments.
Dashboard Visualization	The system shall display worker details, available health information, predicted conditions, and task recommendation details.
Assignment	The system shall store approved task assignments and their

Management	relevant status in MongoDB.
Prediction Information	The system shall display the prediction confidence score when it is provided by the prediction service.

4.1.3 Non-Functional Requirements

Non-functional requirements specify the quality attributes and functional expectations of the software. They include usability, performance, reliability, security, maintainability, and integrity of the data.

Table 4.3: Non-Functional Requirements

Quality Attribute	**Description**
Performance	The system should process physiological measurements and generate predictions within a reasonable response time.
Reliability	The system should handle invalid, incomplete, or unavailable sensor data appropriately.
Usability	The dashboard should present worker information and prediction results in a clear and understandable format.
Security	Access to worker information and management functions should be restricted to authorized users.
Data Integrity	Worker details, physiological records, predictions, and task assignments should be stored consistently.
Maintainability	The system should use a modular structure to simplify troubleshooting, maintenance, and future improvements.
Scalability	The system architecture should support future increases in the number of workers and monitoring records.
Privacy	Worker information and physiological measurements should be handled confidentially.
Compatibility	The web application should operate in a supported modern web

	browser.
Accuracy and Limitations	The system should present model predictions as decision-support information and acknowledge the limitations of the trained model.

4.1.4 User Scenarios

User scenarios describe how stakeholders interact with the system to achieve specific objectives. The main scenario involves an estate manager registering a worker, collecting physiological measurements, obtaining a worker condition prediction, and reviewing a recommended task. The supervisor then reviews and approves the assignment.

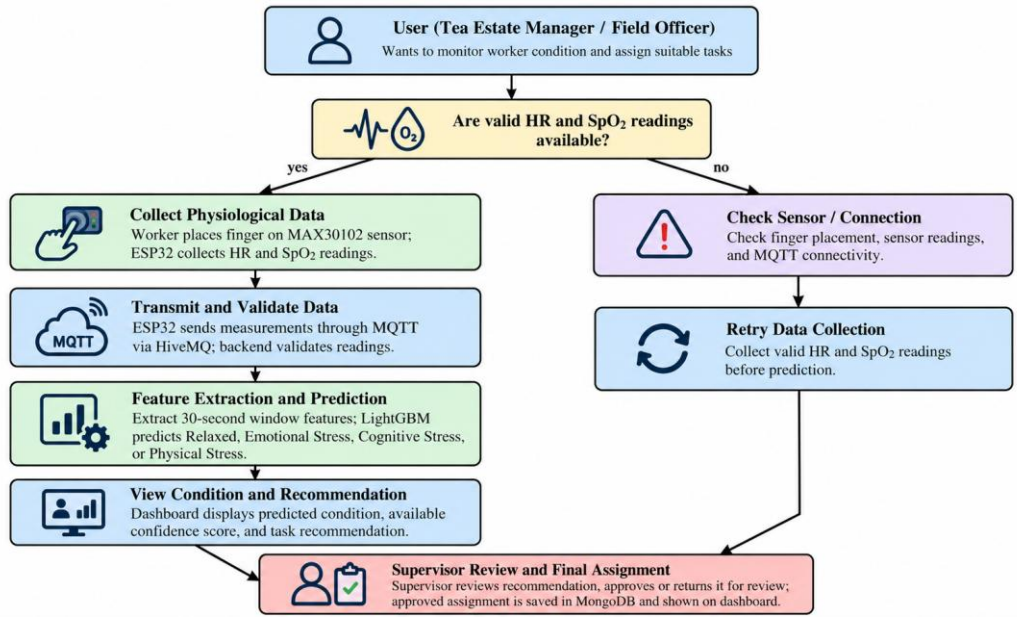


Figure 4.1: User Scenario for the Labor Management and Welfare Optimization System

Table 4.4: User Scenarios

User Scenario	Primary Actor	Expected Outcome
Register a Worker	Estate Manager	The worker's details are recorded in the system.
Collect Physiological Measurements	Tea Estate Worker	HR and SpO_2 measurements are obtained using the IoT sensor.
Process Sensor Data	System	Received measurements are validated and converted into the required statistical features.
Predict Worker Condition	System	The LightGBM model produces one of the four worker condition categories.
Recommend a Task	System	A task recommendation is generated according to the configured risk-management rules.
Review and Approve a Task	Supervisor	The recommendation is reviewed and the assignment is approved or returned for review.
View Worker Monitoring Results	Estate Manager	The dashboard displays available worker information, predictions, and task assignment status.
Store Assignment Information	System	The approved assignment and relevant records are saved in MongoDB.

4.2 Component architecture within the complete solution

Intelligent Decision Support System for Sri Lankan Tea Estate Management is the system that follows client-server architecture pattern using the web platform. The entire system comprises of four main parts: Labor Management and Welfare Optimization, Tea Leaf Disease Recognition and Pattern Detection, Weather and

Climate Intelligence Specific to Tea, and Automated Sorting and Grading of Tea Using AI. Each of these parts serves certain purposes and gives respective information.

The Labor Management and Welfare Optimization part is concerned with the monitoring of workers' physiological state and appropriate task distribution. It involves IoT enabled data gathering, machine learning enabled worker's state prediction, backend service provision, and database management through web platform.

Main architecture components

Frontend (React.js): This component provides an interface for registering workers, accessing physiological monitoring data, accessing the predicted worker condition, and viewing task recommendations.

Backend (Node.js and Express.js): The component receives physiological measurements, validates incoming data, controls application processing, interacts with the prediction model, and deals with database management.

IoT device (ESP32 and MAX30102): The device gathers physiological measurements, HR, and SpO_2 from workers, preparing data for sending it to the backend.

Communication (MQTT and HiveMQ): The communication component sends physiological measurements collected by the IoT device to the backend using the MQTT communication protocol and HiveMQ broker.

Machine Learning model (LightGBM Classifier): This model processes physiological features and predicts worker condition as one of the following: Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress.

Database (MongoDB): The component contains worker information, physiological monitoring data, prediction results, and task assignment data.

Dashboard and task management: Available predictions, confidence score where possible, task recommendations, and task approvals allow estate managers and supervisors to analyze worker conditions and tasks.

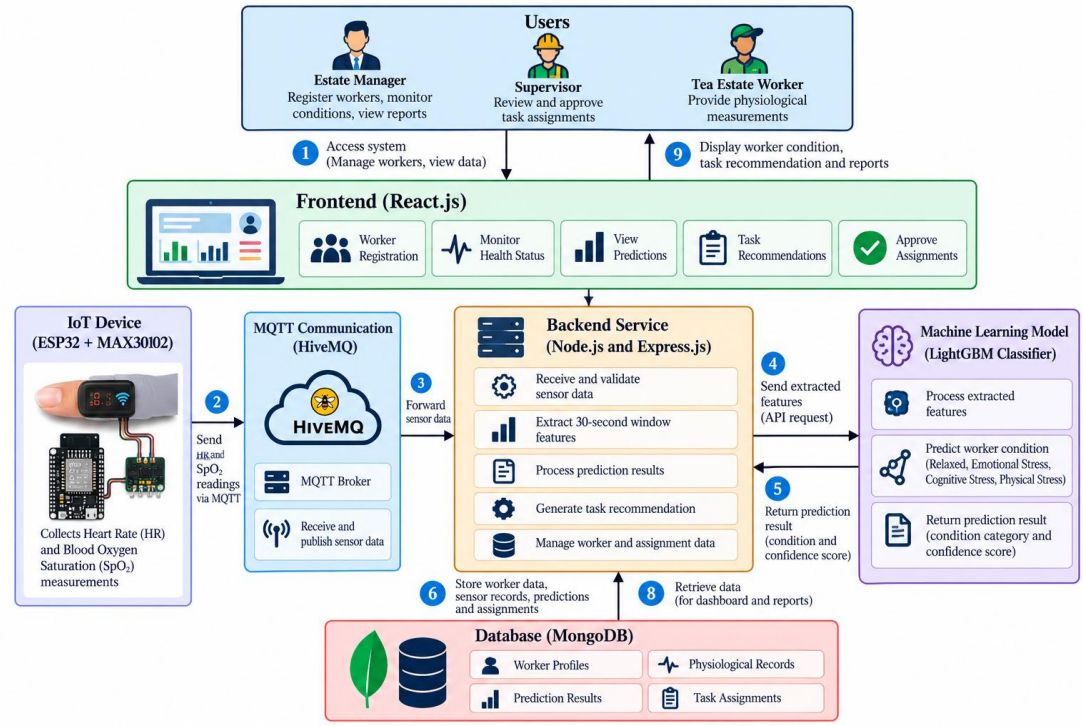


Figure 4.4: Component Architecture of the Labor Management and Welfare Optimization System

4.3 Design alternatives and decisions

Various design choices were evaluated during the development of the Labor Management and Welfare Optimization module. These involved the choice of the algorithm, front-end framework, back-end framework, communication protocol for IoT, and the database. The chosen frameworks help in physiological data acquisition, worker state prediction, task suggestion, and results display in the web application.

Table 4.4: Design Alternatives and Decisions

Design Aspect	Alternative	Selected Decision	Reason
Machine learning model	Other machine learning classification algorithms	LightGBM Classifier	Selected to classify worker conditions using physiological features such as HR and SpO_2.
Frontend	Other frontend	React.js	Supports a component-based interface for

development	frameworks		worker registration, monitoring, and displaying prediction results.
Backend development	Processing application logic entirely in the frontend	Node.js and Express.js	Handles application requests, validates sensor data, and coordinates communication between the frontend, prediction model, and database.
IoT communication	Direct communication without a message broker	MQTT with HiveMQ	Provides a communication mechanism for transmitting physiological measurements from the ESP32 to the backend.
Physiological data collection	Manual recording of worker measurements	ESP32 with MAX30102 sensor	Supports the collection of HR and SpO_2 measurements from the worker.
Database	Relational database	MongoDB	Provides document-based storage for worker profiles, physiological records, predictions, and task assignments.

Worker condition presentation	Displaying only raw physiological measurements	Displaying predicted conditions and relevant monitoring information	Makes the model's output easier for estate managers and supervisors to review.
Task assignment	Assigning tasks without supervisor review	Task recommendations with supervisor approval	Supports human oversight before recommended assignments are finalized.

The LightGBM Classifier was used in implementing the classifier for the categorization of worker physiological conditions using HR and SpO_2. There are four categories, which include Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress. The findings indicated 82% accuracy for the classification of the workers under the conditions of the experiment.

React.js was used in developing the frontend for the reason that its component-based design will facilitate the development of the interactive dashboard. The backend processing is done by Node.js and Express.js to validate the physiological measurement data as well as communicate with the prediction model and database.

As for the IoT communication, MQTT together with HiveMQ was selected for transmitting the measurements collected via ESP32 and MAX30102 sensors. The MongoDB is chosen as the database where the information about the worker, physiological monitoring, prediction results, and tasks will be stored. Finally, the consideration of supervisors' opinion about the task recommendations has been taken into account.

4.4 Detailed design

The detailed design of the Labor Management and Welfare Optimization component describes how the frontend, backend, IoT devices, ML model, and database interact with one another in order to monitor the state of the workers and facilitate task allocation. This involves worker registration and physiological measurements, data

transmission, feature extraction, worker state prediction, task recommendation, and finally approval from the supervisor.

4.4.1 Frontend Design

The frontend is developed using React.js and acts as the graphical user interface for estate managers and supervisors. The frontend enables users to register workers, see the physiological monitoring data, check the predicted worker's condition, and see the recommended tasks.

Some of the functionalities that exist in the frontend include worker registration, showing available heart rate and oxygen saturation levels, displaying predictions, seeing the recommended tasks, and assignment approval status.

4.4.2 Backend Design

The back end is implemented using Node.js and Express.js and handles communication between the front end, IoT data communication module, machine learning model, and database. The back end receives physiological parameters sent over MQTT via the HiveMQ broker and checks for data integrity prior to further processing.

It is also responsible for handling the extraction of features and prediction, processing the received results about the state of the worker, managing task recommendation data, and performing all operations related to the database.

4.4.3 IoT Data Collection and Communication Design

This part of IoT comprises the ESP32 microcontroller connected to the MAX30102 sensor to gather the HR and SpO_2 values. In the process of gathering the data, the employee needs to place his/her finger on the sensor in order to gather the necessary physiological data.

ESP32 sends the gathered values via the MQTT protocol with the help of HiveMQ broker. The backend gathers and validates the data in order to prepare it for prediction. Data is gathered during a 30-second period in order to extract the necessary statistical features needed for ML algorithms.

4.4.4 Machine Learning Model Design

This module predicts the worker's state based on physiological information through the LightGBM Classifier (LGBMClassifier). Training the classifier is done based on the PhysioNet Non-EEG Database for Neurological State Assessment, where HR and SpO_2 are the major physiological input features.

Prior to the prediction, the system extracts a set of statistical features out of the 30-second window. These features are the average, maximum, minimum, and standard deviation of HR and SpO_2 values, and also the count of SpO_2 drops. The features are fed into the trained classifier to predict one of four classes – Relaxation, Emotional Stress, Cognitive Stress, and Physical Stress.

The reported accuracy of this model according to the research results was 82%. The predicted state is sent back to the application and shown on the dashboard.

4.4.5 Database Design

This database will be used for storing the information that is needed for the Labor Management and Welfare Optimization component.

The primary information related to the component is as follows:

- Information about the workers – including their name, designation, and department.
- Information regarding the physiological monitoring data that comprises HR and SpO_2 values.
- Physiological feature values extracted from the physiological monitoring data wherever applicable.
- Information regarding the predicted worker condition and the prediction.
- Task recommendation and approval from the supervisor.
- Approved task assignment information.

The backend fetches the appropriate information based on the requirements of the application. This will allow the dashboard to display the information about the workers, monitoring data, and task assignments.

4.4.6 Task Recommendation and Dashboard Design

Once the worker state has been predicted, the system applies the configured risk-management rules to provide task recommendations. Since the four categories produced by the LightGBM algorithm are different from the risk level in terms of task recommendation, the system needs to have a mapping between the prediction result and the risk level before task recommendation.

The dashboard displays the available worker data, predicted states, confidence level of the prediction, recommended tasks, and the approval of the assignment. Once the supervisor approves the recommendation, the information regarding the assignment is saved in MongoDB. This approach ensures that human intervention is part of the task assignment process.

4.5 Standards and professional practice

Software engineering considerations for Labor Management and Welfare Optimization component design include security concerns, and the responsible application of machine learning.

- API Security: Ensure security of the backend and sensor communications, and safeguard credential details.
- Open-Source Licenses: Adhere to open-source licenses of the libraries, frameworks and datasets utilized.
- Data Protection: Safeguard private data of workers and physiological measurements.
- Responsible AI: Display worker condition predictions as decision support information, not medical diagnosis.
- Human Oversight: Include supervisor approval of task suggestions before making final assignments.
- Data Validation: Validate physiological measurements and maintain accurate worker and task information.
- Academic Citation Style: Cite according to IEEE citation style, as per the requirement of the department.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The Labor Management and Welfare Optimization module is designed based on a web-based system architecture which is combined with Internet of Things technology and a machine learning framework. This platform can be used to conduct the activities of physiological data acquisition, work status prediction, task recommendation, and result presentation.

Table 5.1: Development Environment and Technology Stack

Technology	**Purpose**
React.js	Frontend development and dashboard presentation
Node.js	Backend application development
Express.js	Handling backend routes and application requests
ESP32	Collecting sensor readings and providing Wi-Fi connectivity
MAX30102	Measuring Heart Rate (HR) and Blood Oxygen Saturation (SpO_2)
MQTT	Transmitting physiological measurements
HiveMQ	Receiving and distributing MQTT messages
LightGBM Classifier	Predicting worker conditions from physiological features
MongoDB	Storing worker details, monitoring records, predictions, and task assignments
Google Colab	Machine learning model development and training
Visual Studio Code	Software development environment
Postman	API testing, where applicable

5.2 Component implementation

This section integrates the Physiological Measurement through IoT component with the Machine Learning-based Condition Prediction component. The ESP32 microcontroller acquires HR and SpO2 measurements from the MAX30102 sensor and sends them via MQTT using the HiveMQ broker.

The backend part, written on Node.js and Express.js, receives and validates these measurements. The system analyzes the following statistical features of the readings received during the 30 seconds interval: average, max, min, and standard deviation of HR and SpO2 values and the quantity of SpO2 drops.

These features are classified using the trained LightGBM classifier. It predicts one of the four states of the worker: Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress. This prediction is available via the React.js dashboard that is aimed at the monitoring of the workers and task allocation. The MongoDB database stores the corresponding information about the workers, the results of the monitoring and classification, and tasks allocated.

5.3 Integration with the other components

The Labor Management and Welfare Optimization component is one of the four major components of the Intelligent Decision Support System for Sri Lankan Tea Estate Management. This component has similar web-based architecture as other three components, which are Tea Leaf Disease Detection and Pattern Analysis, Tea-Specific Weather and Climate Intelligence, and Automated Tea Sorting and Grading Using AI.

This component concentrates specifically on worker physiological monitoring and welfare task management. Integration is performed within the main web-based framework using React.js frontend, Node.js and Express.js backend, and MongoDB database. Physiological data is received via IoT communication protocol, whereas LightGBM Classifier predicts worker condition.

The component provides worker monitoring data and task recommendation output to the whole decision support system.

5.4 Verification and software testing

The testing is an essential process for ensuring the Labor Management and Welfare Optimization module meets its required tasks. It needs to include such processes as worker registration, physiological data acquisition, MQTT communication, data validation, feature extraction, worker condition prediction, data base manipulation, and dashboard display.

Table 5.2: Component Test Cases

Test ID	Test Scenario	Expected Result
TC01	Register a new worker	Worker details are accepted and stored in MongoDB.
TC02	Collect HR and SpO_2 measurements using the sensor	Physiological measurements are obtained from the MAX30102 sensor.
TC03	Transmit sensor readings through MQTT	Sensor messages are received through the configured HiveMQ communication process.
TC04	Validate incoming physiological data	Valid readings are accepted for processing, while invalid readings are handled appropriately.
TC05	Extract features from a 30-second measurement window	The required statistical features are generated from the collected readings.
TC06	Submit extracted features to the LightGBM Classifier	The model returns a worker condition prediction when processing succeeds.
TC07	Display prediction results on the dashboard	The predicted worker condition is presented to the user.
TC08	Generate a task recommendation	A recommendation is produced according to the configured risk-management rules.

TC09	Review and approve a task assignment	The supervisor can review the recommendation and approve or return it for further review.
TC10	Store an approved task assignment	Assignment information and relevant status are saved in MongoDB.
TC11	Display worker monitoring information	The dashboard presents available worker details, prediction results, and assignment status.
TC12	Process missing or invalid sensor data	The system handles the input appropriately without treating invalid readings as reliable predictions.

5.5 Deployment and operation

The Labor Management and Welfare Optimization component runs via the web application that is connected to the IoT sensor monitoring device, communication protocol MQTT, backend services, ML model, and MongoDB database.

At runtime, the MAX30102 sensor detects HR and SpO_2 values, while the ESP32 sends the data via MQTT protocol using HiveMQ service. The backend processes the measurements received and creates features needed by the LightGBM Classifier. The result of the worker health state prediction is visualized on the dashboard.

After reviewing the recommended task, the supervisor approves the task assignment, and the corresponding assignment details are saved in MongoDB. The operation of the component depends on the availability of the configured backend services, database connection, communication mechanism, and prediction model.

Proper configuration, credential management, error handling, and monitoring of services are important in order to ensure reliable operation. The accuracy of the model equal to 82% was reported and is relevant only to the research evaluation and does not guarantee the same results in each tea estate.

6 RESULTS AND VALIDATION

6.1 Research results

The trained machine learning model was validated by using a test set consisting of 1,279 samples based on subjects. The accuracy of the trained machine learning model was found to be 87.33% and the balanced accuracy is 84.49%. The precision of the machine learning model is 84.13% (macro), recall of the machine learning model is 84.49% (macro), and the F1-score of the machine learning model is 84.28% (macro). The weighted F1-score was found to be 87.50%.

The classification report is done for the three classes namely, AEROBIC, ANAEROBIC, and STRESS. For the STRESS class, the precision is 100.00%, recall is 96.44%, and the F1-score is 98.19%. The AEROBIC class had a precision of 80.65%, recall is 83.14%, and F1-score is 81.87%. For the ANAEROBIC class, the precision is 71.74%, recall is 73.88%, and the F1-score is 72.79%.

From the confusion matrix, it is observed that out of 421 AEROBIC samples, 350 samples were accurately classified and 71 were misclassified as ANAEROBIC samples. Out of 268 ANAEROBIC samples, 198 were accurately classified and 70 were misclassified as AEROBIC samples. For the 590 STRESS samples, 569 samples were accurately classified as STRESS samples, 14 samples were misclassified as AEROBIC samples, and 7 samples were misclassified as ANAEROBIC samples.

From the results, it is clear that the model performed the best for the classification of the STRESS samples among the three classes. The lower F1-score for the ANAEROBIC samples suggests that the separation of ANAEROBIC samples from the AEROBIC samples is still difficult.

Table 6.1: Fine-Tuned Model Evaluation Results

Evaluation Metric	Result
Accuracy	87.33%
Balanced Accuracy	84.49%

Macro Precision	84.13%
Macro Recall	84.49%
Macro F1-score	84.28%
Weighted F1-score	87.50%
Total test samples	1,279

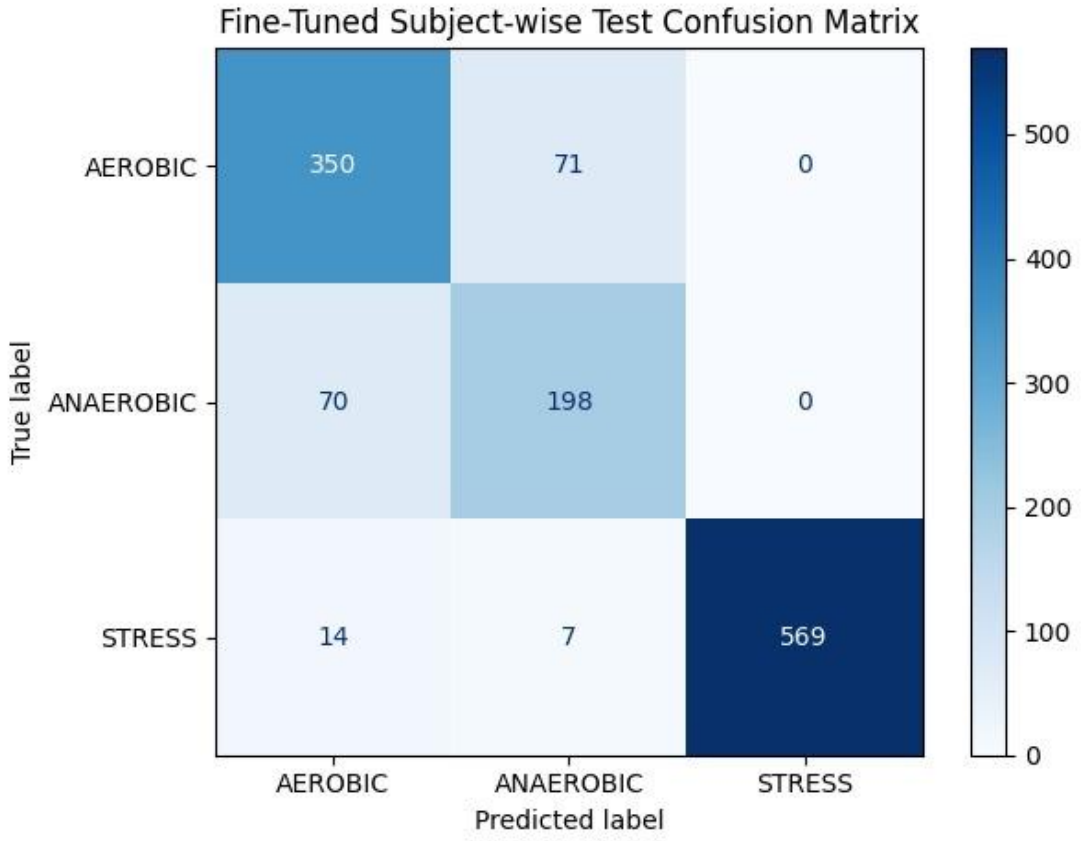


Figure 6.3: Confusion Matrix of the Fine-Tuned Model on the Subject-Wise Test Dataset

```
Classification report:

              precision    recall  f1-score   support

    AEROBIC     0.8065    0.8314    0.8187       421
  ANAEROBIC     0.7174    0.7388    0.7279       268
     STRESS     1.0000    0.9644    0.9819       590

   accuracy                         0.8733      1279
  macro avg     0.8413    0.8449    0.8428      1279
weighted avg     0.8771    0.8733    0.8750      1279
```

Figure 6.2: Classification Report of the Fine-Tuned Model

```
======================================================
FINE-TUNED MODEL — SUBJECT-WISE TEST RESULTS
======================================================
Accuracy:          87.33%
Balanced Accuracy: 84.49%
Macro Precision:   0.8413
Macro Recall:      0.8449
Macro F1:          0.8428
Weighted F1:       0.8750
```

Figure 6.3: Confusion Matrix of the Fine-Tuned Model on the Subject-Wise Test Dataset

6.2 Product and technical results

Labor Management and Welfare Optimization module utilizes IoT based physiological monitoring along with the machine learning model and the web application for assisting in worker condition monitoring and task management. The technologies used in this module are React.js for the frontend, Node.js and Express.js for backend, ESP32 and MAX30102 for collecting physiological measurement, MQTT with HiveMQ for communicating measurement data, LightGBM Classifier for predicting worker condition and MongoDB for storing the related information.

Major Technical functionalities include -

- Obtaining physiological measurements such as Heart Rate (HR) and Blood Oxygen Saturation (SpO_2) through the IoT sensor.
- Communicating physiological measurements via MQTT with HiveMQ.
- Extraction of statistical features from the sensor measurements gathered within 30 seconds window.
- Predicting the worker condition using the LightGBM Classifier.
- Categorizing the worker condition into Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress.
- Generating the tasks as per pre-defined risk management rule.
- Task Assignment Review and Approval with the involvement of supervisor.
- Storage of worker profile, physiological monitoring record, prediction result, and task assignment in MongoDB.
- Display of worker condition prediction and task recommendation information on React.js dashboard.

6.3 User or stakeholder validation

The Labor Management and Welfare Optimization component is designed to support estate managers and supervisors in monitoring worker conditions and managing task assignments. The dashboard presents worker information, physiological measurements, predicted worker conditions, and task recommendations to assist users in making informed decisions.

The main areas considered for stakeholder validation include:

- **Usability:** Evaluating whether users can understand and navigate the dashboard.
- **Information Presentation:** Checking the clarity of physiological measurements and worker condition predictions.
- **Task Recommendations:** Assessing whether the recommended tasks are presented clearly for supervisor review.
- **Approval Process:** Verifying whether supervisors can review and approve task assignments.

- **Data Accessibility:** Checking whether relevant worker information and assignment records can be accessed when required.

Stakeholder feedback can help identify usability issues and improvements needed to make the component more suitable for tea estate management. Actual user validation results should be documented based on feedback collected from estate managers, supervisors, or other intended users.

6.4 Integration results

The component architecture encompasses the React.js Front End, Node.js/Express.js Back End, IoT Monitoring Devices, LightGBM Classifier, and the MongoDB Database. The planned workflow will include gathering HR and SpO_2 measures using the ESP32 and MAX30102 sensor, sending the information via MQTT using HiveMQ, extracting physiological features, predicting the status of workers, and showing the information via the dashboard.

This component architecture links physiological data acquisition with worker status prediction and task recommendations. The MongoDB Database will be used for storing information about workers, their monitoring, prediction outcomes, and task assignment data. Supervisors will look at the recommended tasks before task assignment approval.

The component architecture and workflows are illustrated in the following diagrams, but actual test records that cover sensor data transmission, feature extraction, predictive models, database operations, and dashboard presentation are required for confirmation of the end-to-end integration. In this regard, the integration outcome should be backed up by test results of the developed system.

6.5 Summary of findings

It can be deduced from the results obtained that the Labor Management and Welfare Optimization module utilizes physiological monitoring through the Internet of Things, machine learning, and web technologies to promote the welfare of the workers and manage tasks in the tea estates. In the process, the HR and SpO_2

measurements are gathered by the use of the ESP32 and MAX30102 sensors and then transmitted using the MQTT protocol via the HiveMQ broker.

The measurement values are further analyzed to get the features related to the health conditions of the workers. The LightGBM Classifier is used to predict the worker health condition based on physiological features. The output of the model enables recommendation of tasks based on predetermined risk management rules. The React.js dashboard displays worker details, predicted health condition, recommended tasks, and their approval status, while Node.js and Express.js takes care of the processing and communication with the MongoDB database.

7 DISCUSSION

7.1 Interpretation of findings

The results show that the use of IoT technology along with machine learning can help monitor the state of workers and their task management in tea estates. The HR and SpO_2 are measured through the ESP32 and MAX30102 sensors. The data is sent using MQTT protocol and HiveMQ as the broker, and then, it is processed on the backend where the physiological features are derived.

The prediction of worker states is done using the LightGBM Classifier based on the derived features. The predictions enable task recommendations according to the defined risk management rules. The dashboard shows the worker details, their predicted states, recommended tasks, and whether the recommendation is approved or not.

According to the provided results of evaluation of the fine-tuned model, the obtained values of accuracy, balanced accuracy, and macro F1-score equal 87.33%, 84.49%, and 84.28% correspondingly and are based on 1,279 test instances. It is evident that the evaluated model demonstrated a rather high performance level concerning its classification task at the test dataset. Still, according to the reported classes (AEROBIC, ANAEROBIC, and STRESS), these results can be considered valid if the model being evaluated and the one used in the implemented system is proven to be identical. The model needs to be tested using data obtained from tea estate workers.

7.2 Comparison with literature baselines and existing solutions

Current approaches to worker management include human observation, logging of work-related information, and decisions by a supervisor. Even though these approaches can be used to manage workers daily, physiological monitoring and condition prediction might be beneficial in providing some additional information regarding the worker's well-being and task allocation.

The proposed module will utilize physiological data collection, condition prediction, and task recommendation in a web-based application. While current solutions include only information logging, the proposed system will be capable of processing HR and SpO_2 measurements and presenting predicted conditions to a supervisor. The use of MQTT protocols and the HiveMQ broker will facilitate the transferring of sensor information to the backend server, while the MongoDB database will store worker, monitoring, prediction, and assignment data.

Accuracy reported in the research paper for the LightGBM-based worker condition prediction model within the overall TeaGuard system is 82%. Fine-tuned model evaluation results show that the accuracy is 87.33%. These accuracies are not directly comparable unless it is determined that both models, datasets, classes, and evaluation procedure are comparable. Nevertheless, these results highlight the significance of performing classification model evaluation on well-defined datasets.

7.3 Research contribution

The first research contribution of this component is the development of the methodology for integrating physiological monitoring and worker condition predictions based on machine learning. The HR and SpO_2 values are transformed into statistical features that allow classifying the worker condition with the help of a classification algorithm.

The second contribution is the relation between the prediction results and the task recommendation. In contrast to just showing the physiological measurements, the suggested approach allows utilizing the predictions to make decisions about the right work tasks and recommendations for taking rests.

The component also paves the way for the future research on using technologies to ensure workers' well-being in the tea estates of Sri Lanka. For example, the approach can be evaluated with the help of real data from tea estates, different machine learning models can be compared, and the use of physiological monitoring to improve worker safety while not substituting expert decision making can be researched.

7.4 Product contribution

Product contribution

This is a web-based Labor Management and Welfare Optimization System that has IoT data collection, backend processing, machine learning, database storage and dashboard based monitoring. React.js will be used for building the frontend while Node.js and Express.js will be used for backend operations. ESP32 and MAX30102 sensor will be used for collecting physiological measurements and for communication, MQTT together with HiveMQ will be used.

System design entails presentation of information about workers, physiological measurements, predictions, confidence scores if available, recommended tasks and approval status of the assignments. MongoDB will provide storage for the relevant information about workers, monitoring, predictions and assignment of tasks.

The supervisor review is a major product feature. The recommended tasks will be reviewed instead of accepting them as automatic decisions. This is aimed at ensuring that human intervention is allowed when managing the workforce. The component can thus act as a basis for workforce monitoring and task allocation in tea estates.

7.5 Limitations and threats to validity

Some of the limitations that need to be considered when analyzing the results include the following. Firstly, the performance of the machine learning algorithm is dependent on the quality and appropriateness of the data used for training the algorithm. Despite the usage of PhysioNet dataset for training, the data in the dataset may not adequately represent the physiological factors, working pattern, environmental factors, and demographics of the tea estate workers in Sri Lanka. This implies that the performance of the algorithm using the training or testing datasets will not reflect its performance in the actual tea estate.

Secondly, both HR and SpO_2 levels could be influenced by factors such as sensor positioning, movement, environmental factors, and quality of the measurements. The 30-second window used for the measurement of the readings would have some bearing on the predictions made.

Thirdly, the mere fact of good classification does not show that this system leads to increased welfare of the workers or appropriate assignment of tasks to them. This requires practical evaluation, comparison with alternatives, and feedback from workers and managers. Moreover, the metrics of the reported fine-tuned models should be clearly separated from the metrics of the LightGBM model in case no confirmation of their connection is provided.

Finally, the recommendations on assigning tasks proposed by this system depend on predetermined risk management policies. These policies need to be carefully evaluated to make sure that they are correct for various kinds of work and situations. It should be remembered that this system is only a recommendation one and is not a medical diagnostic system.

7.6 Implications

There is a possibility for this component to have certain implications for worker welfare and workforce management in the Sri Lankan tea industry. Through a combination of physiological measurements and machine learning predictions, this component could assist management and supervisors in gaining more insight regarding worker conditions and task allocation.

With respect to tea estate management, the dashboard and task recommendation capabilities of the component could assist in more systematic monitoring and documentation. With regards to the documentation, any patterns that require further analysis could be identified by management, assuming the accuracy and proper handling of the data.

In terms of technology, the use of IoT devices, MQTT communication, machine learning, and web-based dashboard represents a potential architecture for data-driven worker management. Nonetheless, there must be reliable sensor readings, proper data handling, clear decision-making guidelines, and adequate human oversight.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The Labor Management and Welfare Optimization System will assist tea estate managers and supervisors in monitoring the status of workers and allocating tasks to them. The system will collect HR and SpO_2 values of workers using ESP32 and MAX30102 sensors. These values will then be used as inputs into a LightGBM Classifier for predicting the worker status using defined risk management rules.

The main target users will be small, medium, and large tea estates, plantation managers, and field supervisors in Sri Lanka. With the inclusion of physiological monitoring and a web interface, the proposed component may potentially provide better access to information on worker status and allow better task allocation decisions. However, it needs to be tested in practice.

8.2 Market and Competitive Position

Tea plantations play a significant role in the agriculture industry in Sri Lanka; hence, proper workforce management is crucial to maintain day-to-day operations. It is possible that conventional methods of monitoring workers and allocation of tasks may rely on manual observations and supervision. This suggested module represents an alternative to this method by incorporating physiological monitoring through IoT, machine learning prediction, and task recommendation using the web application.

One way that this could become a competitive advantage is through the consolidation of HR and SpO_2 data gathering, worker conditions prediction, task recommendation, and supervisor approval in one single system. Use of MQTT through HiveMQ ensures communication between the IoT device and the backend, whereas the dashboard allows access to the necessary information about the workers and their assignments. However, its competitive value would still have to be evaluated based on the current solutions for workforce management and practical testing.

8.3 Business or Sustainability Model

One option is for the Labor Management and Welfare Optimization System to be provided as a part of a subscription-based tea estate management service. It could be

possible to charge the tea estates based on the number of registered workers, the number of monitoring devices connected, or the features included in the service. Further revenue sources can include the cost of installing the system, training staff, and providing support and maintenance.

For long-term viability, the software needs to be maintained, the sensors need to be monitored, and the machine learning model needs to be assessed periodically. In addition, it could need to be updated based on new data, especially the data representing workers from Sri Lankan tea estates. Collaboration with tea estate owners, representatives of the labor force, and experts on health and agriculture might also enhance its applicability.

The presented business models are hypothetical possibilities but not real-life business operations. Their viability must be assessed by market research and other methods.

8.4 Scalability, Intellectual Property and Compliance

The functionality of this component can be further expanded to support other tea plantations by enhancing the infrastructure of the web application, connecting more IoT devices, and improving operations at the backend and database level. Enhancements can be added in terms of mobile functionality, worker monitoring capabilities, and improved reporting functionalities. Also, the prediction model should be tested on representative data from different tea-producing regions to see how consistent its performance is under varying working conditions.

In terms of intellectual property rights, the project team should take care of copyright issues related to the source code of the project, license requirements of the used software libraries and machine learning algorithms, as well as the acceptable use of the training set. When deploying the project commercially, it will be necessary to ensure that all the worker data is protected.

Since this component evaluates health-related measurements and issues task recommendations, its predictions should be considered support information and not medical diagnoses. It would be important to keep the responsibility for reviewing recommendations and get professional advice when needed. Compliance with any

data protection rules and proper work safety practices should be checked prior to any commercial release.

8.5 Contribution to the Sustainable Development Goals

There are several possible SDGs to which the Labor Management and Welfare Optimization System might have relevance.

SDG 3 – Good Health and Well-being: The component is created to track the HR and SpO_2 measurements of workers and give predictions on the working condition that might increase the awareness of workers' welfare. The contribution of the component to good health would have to be verified through practical testing.

SDG 8 – Decent Work and Economic Growth: The use of the component in terms of workforce management by giving data about task recommendations and the possibility to check assignments might lead to better consideration of workers' welfare and proper assignment of the work.

SDG 9 – Industry, Innovation, and Infrastructure: The component uses IoT technology, machine learning, and web development to make innovations in the field of managing the tea estates of Sri Lanka.

SDG 12 – Responsible Consumption and Production: Better task planning and workforce management might help to increase efficiency in managing the estate. However, it would have to be proved practically.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

Labor Management and Welfare Optimization System has been built as part of the Intelligent Decision Support System for Sri Lankan Tea Estate Management. The main goal of the solution is worker condition monitoring and management with the use of the Internet of Things (IoT), machine learning and web development technologies. The system uses the ESP32 and MAX30102 sensors to acquire HR and SpO_2 measurements of workers.

The acquired measurements are transferred with the help of MQTT and HiveMQ Broker to be further analyzed by the backend. The statistical features are being extracted from 30 seconds of measurements and the predictions of worker condition are made with the help of the LightGBM Classifier. The system will display predictions and suggestions on tasks via the React.js dashboard, with the use of Node.js, Express.js in the backend and MongoDB for worker, monitoring and assignment data storage.

One significant aspect of this component is that there is a relationship between predicting worker condition and recommending tasks. The supervisor can examine the recommendation before assigning any task to workers, thereby ensuring that human element also forms a part of decision making. In all, this component forms the basis of technical infrastructure for worker monitoring and workforce management in tea estates. Nevertheless, more testing is required to see how effective this component is in practice.

9.2 Achievement of objectives

The component achieves its primary goals by combining physiological data gathering, machine learning prediction and task management capabilities.

The first goal is to gather physiological data of workers using IoT technology. The suggested implementation is based on the use of the MAX30102 sensor together with the ESP32 to gather HR and SpO_2 data and send it via MQTT using HiveMQ.

The second goal is to prepare the gathered data for further machine learning. The window for data gathering is set at 30 seconds with the following parameters calculated for this period: mean, maximum, minimum and standard deviation values for HR and SpO_2 as well as the amount of SpO_2 drops.

The third goal is to predict worker conditions using machine learning. The LightGBM Classifier is used to predict condition categories introduced into the model: Relaxed, Emotional Stress, Cognitive Stress and Physical Stress. The performance of the implemented model should be evaluated by using evaluation results for this particular model and test dataset.

Fourthly, the objective was to facilitate worker task allocation using predictions. This module is expected to map worker condition predictions to specified risk management rules and give recommendations for tasks for consideration by the supervisor.

Finally, the objective was to offer relevant data presentation using a dashboard and keep relevant records. React.js for the frontend, Node.js, and Express.js for the backend, along with the MongoDB database will make up the architecture of data presentation and record-keeping of worker and assignment data.

The above objectives have set out the functionality required from this component. The degree to which these objectives have been met should be assessed using implementation data and testing evidence.

9.3 Individual research and product contributions

Individual research contribution is about the application of machine learning and IoT-based physiological monitoring for the purposes of managing the worker wellbeing and tasks in tea estates of Sri Lanka. The component examines ways to process HR and SpO_2 measures in order to obtain the valuable features and make predictions about workers' state. Furthermore, the component examines how the predictions can be used to provide recommendations regarding tasks while still ensuring that supervisors retain their decision-making authority.

Individual product contribution is the design and development of Labor Management and Worker Welfare Optimization System implemented with the help of React.js, Node.js, Express.js, MongoDB, ESP32, MAX30102 sensor, MQTT, HiveMQ, and LightGBM Classifier. The component combines data collecting, communication, feature extraction, condition predictions, dashboards, and tasks recommendations in one system.

The combination of the technologies described above serves as a basis for developing the solutions for managing tea estates workers in the future. The component also contributes to the TeaGuard research project in general since it tackles the problem of the workforce and its wellbeing.

9.4 Future directions

There are several ways that could be implemented to improve the component.

First, further analysis should include physiological data obtained from tea estate workers while they work. It will allow checking whether the training dataset is a good representation of the target population and whether the predictions made by the model are reliable under various working conditions.

Second, the machine learning model can be improved by providing more representative data, analyzing features and comparing other classification algorithms. The effectiveness of the model needs to be estimated using metrics such as accuracy, balanced accuracy, precision, recall and F1-score. Updated model needs to be tested separately before implementing into the application.

Third, IoT data collection process needs to be optimized by analyzing reliability of sensors, handling of missing or incorrect data and testing communication via MQTT on different network conditions.

Fourth, the rules for task recommendations can be refined based on input from tea estate managers, employees, and experts in health and occupational safety. The recommendations must be assessed to confirm their relevance to different work tasks and avoid considering machine learning predictions to be medical diagnoses. Human

judgment must continue to be an important element in decision-making that might have consequences for worker safety.

Lastly, the component can be enhanced by providing mobile-friendly access, better reporting, trend analysis, and scalability to handle more workers and monitoring devices. Future usability testing and piloting of the component can provide feedback on whether the component is useful for tea estates and what further enhancements it needs before it can be implemented widely.

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GLOSSARY

Feature extraction - The step when raw physiological measurements are transformed into numerical features usable for machine learning purposes.

Gradient boosting - A technique of machine learning based on the creation of an ensemble of decision trees for prediction.

Human oversight - The participation of a responsible person in checking recommendations made by the system and making the final decision.

IoT (Internet of Things) Device - Physical device that gathers data and transfers it through a network.

Machine learning model - Computational model trained on data samples and capable of identifying patterns and predicting outcomes.

Physiological monitoring - Gathering and observing the body parameters, like heart rate and blood oxygen saturation.

Prediction confidence score - Numerical estimate of model's certainty about the predicted outcome, if applicable to the implementation of the system.

Real-time monitoring - Gathering and processing of data on-the-go, depending on communication and computation delays of the system.

Statistical features- Numerical estimates of collected measurements, like average, max, min, and standard deviation.

Task recommendation - Work assignment recommended based on predictions of worker conditions and management rules.

Wearable sensor - Sensor measuring data from or about a person wearing the sensor.

APPENDICES

Appendix A Research instruments or extended results

Supplementary Information Concerning the Development and Evaluation of Labor Management and Welfare Optimization System is contained in this appendix. It gives extra information about the collection of physiological data, evaluation of machine learning, and software testing.

A.1 Physiological Data Collection

The system involves the use of an ESP32 controller and a MAX30102 sensor to capture HR and SpO_2 values. The worker puts his/her finger on the sensor, and the measurements are captured for 30 seconds.

The measurements captured are sent via MQTT via HiveMQ broker. The backend captures the incoming data and extracts the statistical features for machine learning predictions.

- Average, max, min, and standard deviation of HR
- Average, max, min, and standard deviation of SpO_2.
- Number of SpO_2 drops in the measurement period.

A.2 Machine Learning Evaluation Results

The results for the individually provided fine-tuned model evaluation were as follows

Evaluation metric	Result
Accuracy	87.33%
Balanced accuracy	84.49%
Macro precision	84.13%
Macro recall	84.49%
Macro F1-score	84.28%
Weighted F1-score	87.50%

A.3 Additional Software Test Cases

Test ID	Test description	Expected result	Actual result
AT01	Register a worker	Worker information is recorded correctly	To be completed
AT02	Collect HR and SpO_2 readings	Sensor provides measurement data	To be completed
AT03	Transmit data through MQTT	Data reaches the backend through the configured broker	To be completed
AT04	Validate received measurements	Valid data are accepted and invalid data are handled appropriately	To be completed
AT05	Extract statistical features	Required features are calculated from the measurement window	To be completed
AT06	Generate a model prediction	The configured model returns a prediction	To be completed
AT07	Display prediction results	Results appear correctly on the dashboard	To be completed
AT08	Generate task recommendation	A recommendation is produced according to configured rules	To be completed
AT09	Review a recommendation	Supervisor can review the proposed assignment	To be completed
AT10	Save an approved assignment	Assignment information is stored correctly	To be completed

Appendix B: Academic Integrity and AI Use Disclosure

B.1 Purpose of AI Use

AI-based programs were utilized as an auxiliary help for the preparation of the thesis. The usage of such instruments was helpful in the process of academic writing, in structuring technical explanations, and in increasing clarity. However, the research design, implementation procedure, results of experiments, and academic assertions should be checked against the actual project.

B.2 Tools Used

Tool	**Purpose**
ChatGPT	Assistance with organizing thesis sections, improving academic wording, explaining technical concepts, and preparing preliminary descriptions of the system.
Google Colab	Training and evaluating machine learning models, where applicable to the documented experiments.
Visual Studio Code	Software development and code editing.
Postman	API testing, where used during implementation.
Git/GitHub	Source code version control and project collaboration, where used.